Bison-Prophet: A Triple-Stage Hardware-Accelerated Diagnostic Framework for EV Power Systems

Prognostic Health Management for Electric Vehicles
Using FPGA-Based Protection and Bayesian Inference

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Abstract

Wide-bandgap semiconductors have fundamentally changed what is possible in electric vehicle power electronics, but they have also introduced a protection problem that the industry has not yet solved convincingly. Conventional diode-based fuses simply cannot react fast enough to shield Gallium Nitride (GaN) and Germanium devices from the high-voltage transients that can destroy them in under a nanosecond ($\sim 10^{-9}$ s). This paper introduces Bison-Prophet, a Prognostic Health Management (PHM) framework built around three tightly coupled simulation stages: a physics-based data generator, an FPGA-speed threshold controller, and a Bayesian diagnostic engine that we call the *Spike Prophecy* module. Together, these stages move protection from reactive to predictive — flagging impending failures before the hardware ever trips. All high-fidelity simulations were carried out using the NXP S32G274A vehicle network processor as the reference platform.

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1 Introduction

The shift toward wide-bandgap semiconductors in EV power systems has been driven by one straightforward advantage: GaN and Germanium devices switch faster and waste less energy than silicon. But that same property — extreme switching speed — makes them uniquely vulnerable. Unlike silicon MOSFETs, which can absorb a transient spike for several microseconds before failing, GaN devices can reach their breakdown voltage in less than a nanosecond. By the time a microcontroller interrupt fires, or a traditional fuse melts, the damage is already done.

This is not a marginal engineering problem. In automotive environments, voltage transients from inductive loads, regenerative braking events, and ground faults are routine. The question is not whether spikes will occur, but whether the protection system can outrun them. Bison-Prophet was built around that constraint. The core idea is to push the first line of defence into the FPGA fabric itself — where decisions happen in clock cycles rather than milliseconds — and then layer a probabilistic diagnostic model on top to catch failure patterns before they escalate into hardware damage. The result is a framework that is simultaneously fast enough to protect and smart enough to predict.

2 System Architecture Overview

Bison-Prophet is structured as three sequential simulation phases, each built as a standalone MATLAB/Simulink model. The decision to keep the phases separate was deliberate — it makes the pipeline easier to validate, swap out, or extend independently. Phase 1 generates the raw signal data; Phase 2 handles real-time hardware protection; Phase 3 does the diagnostic reasoning. A bridging dataset (`final_parameters_data.csv`) connects the output of Phase 1 to the input of

Phase 3, and the trained Bayesian weights are stored in `BisonBrain_Optimized.mat`. Table 1 summarises the repository structure.

Table 1: Repository File Overview

File	Phase	Function
`plant.slx`	Phase 1	Synthetic data generation (AC/DC conversion)
`evs.slx`	Phase 2	FPGA block simulation (threshold & trip timing)
`triping.slx`	Phase 3	ML diagnostic interface (pattern analysis)
`final_parameters_data.csv`	Bridge	Dataset linking Phase 1 and Phase 3
`BisonBrain_Optimized.mat`	Phase 3	Trained Bayesian network weights

3 Triple-Stage Simulation Workflow

3.1 Phase 1: Synthetic Data Generation (`plant.slx`)

Before any intelligent diagnosis can happen, we need a reliable picture of what the power environment actually looks like under both healthy and stressed conditions. Phase 1 builds that picture. The `plant.slx` model simulates AC-to-DC conversion and extracts three key features that the Bayesian engine in Phase 3 will later learn from: baseline current behaviour, harmonic noise content, and thermal accumulation patterns.

3.1.1 Ohmic Power Relation

A healthy load should draw current in proportion to the applied voltage. This is simply Ohm's Law, but it serves as the baseline against which anomalies are detected:

$$I = \frac{V}{R} \tag{1}$$

where V is the supply voltage and R is the load resistance. Deviations from this expected current draw are one of the earliest signs that something is going wrong.

3.1.2 Total Harmonic Distortion (THD)

Real power systems are never perfectly sinusoidal. Harmonic distortion in the Capacitor Bank Alpha is tracked using the standard THD metric, which gives a single number summarising how much noise has crept into the signal relative to the fundamental:

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + \cdots + V_n^2}}{V_1} \tag{2}$$

Here V_n is the RMS voltage of the n-th harmonic and V_1 is the fundamental. Rising THD is treated as an early failure indicator, particularly for capacitor degradation.

3.1.3 Thermal Accumulation

Voltage spikes do not just cause electrical stress — they generate heat. The thermal accumulation model tracks temperature rise across Medium and High spike events using a discrete-time integral that reflects how heat builds up when dissipation is faster than cooling:

$$T(k) = T(k-1) + \int \frac{P_{\text{dissipated}}}{C_{\text{thermal}}}\, dt \tag{3}$$

$P_{\text{dissipated}}$ is the instantaneous power loss and C_{thermal} is the thermal capacitance of the component. Sustained thermal accumulation without recovery is a strong predictor of imminent failure.

3.2 Phase 2: FPGA Threshold Control (`evs.slx`)

Phase 2 is where speed becomes the primary concern. The `evs.slx` model translates the continuous voltage signal into binary hardware logic and calculates exactly how many clock cycles the system has before a dangerous transient causes irreversible damage. The FPGA does not interpret or reason — it simply compares and acts, which is precisely what makes it effective here.

3.2.1 Digital Comparator Logic

The fundamental decision rule is straightforward — if the input voltage meets or exceeds the threshold, the trip signal goes high:

$$\text{Trip} = \begin{cases} 1 & \text{if } V_{\text{in}} \geq V_{\text{threshold}} \\ 0 & \text{if } V_{\text{in}} < V_{\text{threshold}} \end{cases} \tag{4}$$

The simplicity of this logic is intentional. In hardware, complexity costs time, and time is the one resource we cannot afford to waste here.

3.2.2 High-Speed Trip Timing

Rather than triggering on a single sample — which risks false positives from measurement noise — the system counts how many consecutive samples exceed the threshold before firing. The trip time is then:

$$t_{\text{trip}} = \frac{N_{\text{samples above threshold}}}{f_{\text{sampling}}} \tag{5}$$

where $N_{\text{samples above threshold}}$ is the consecutive overThreshold count and f_{sampling} is the FPGA clock rate. This gives the designer a tunable trade-off between reaction speed and noise immunity.

3.2.3 FPGA Quantization

Floating-point arithmetic does not exist in FPGA fabric. All real-valued signals must be mapped to fixed-point integers before the hardware can process them:

$$V_{\text{fixed-point}} = \lfloor V_{\text{real}} \cdot 2^n \rfloor \tag{6}$$

where n is the number of fractional bits. Choosing n carefully is important — too few bits and the resolution is insufficient to distinguish a real spike from quantization error; too many and the logic becomes unnecessarily large.
A note on speed: Running at over 10^7 cycles per second, the FPGA operates three to four orders of magnitude faster than any software interrupt. This is what separates it from microcontroller-based protection schemes, and why it is the right tool for guarding GaN devices.

3.3 Phase 3: The Intelligence Engine (`triping.slx`)

If Phase 2 is the reflex, Phase 3 is the reasoning. Once signal patterns have been captured and a trip event has been logged, `triping.slx` asks two questions: *what is failing?* and *how much time is left?* The answers come from a trained Bayesian network and a slope-based time-to-failure estimator that together form what we call the Spike Prophecy.

3.3.1 Bayesian Posterior Probability

Not all voltage anomalies look the same. A MOSFET gate failure produces a different signal fingerprint than a degraded capacitor. The Bayesian engine exploits these differences by computing the posterior probability of a specific fault class given the observed pattern:

$$P(\text{Error} \mid \text{Pattern}) = \frac{P(\text{Pattern} \mid \text{Error}) \cdot P(\text{Error})}{P(\text{Pattern})} \tag{7}$$

The prior $P(\text{Error})$ is initialised from historical failure rate data, and the likelihood $P(\text{Pattern} \mid \text{Error})$ is learned from the synthetic dataset generated in Phase 1. In practice, this lets the system point at a specific component rather than just raising a generic fault flag.

3.3.2 The Spike Prophecy (Time-to-Failure)

Even before the voltage reaches the 40 V trip limit, the system can estimate how long it has. By tracking the average rate at which voltage is climbing over a 500-sample buffer, we can project forward to the failure threshold:

$$t_{\text{prophecy}} = \frac{V_{\text{limit}} - V_{\text{current}}}{\left.\dfrac{dV}{dt}\right|_{\text{avg}}} \tag{8}$$

where $V_{\text{limit}} = 40\,\text{V}$ is the hardware trip point and $\left.\dfrac{dV}{dt}\right|_{\text{avg}}$ is the mean voltage slope. This estimate is deliberately conservative — the 500-sample window smooths out noise while remaining responsive to genuine degradation trends.

3.3.3 System Health Score

Finally, all of this feeds into a single dashboard metric that non-specialist operators can act on:

$$\text{Health}\,\% = \big(1 - P(\text{High_Spike})\big) \times 100 \tag{9}$$

A score of 100% means the system sees no credible threat. As the probability of a high-severity spike climbs, the health score drops proportionally. When it crosses a configurable warning threshold, the dashboard raises the Spike Prophecy alert.

4 Table of Symbols

Table 2: Symbol Definitions

Symbol	Description	Unit
V	Supply voltage	V
I	Load current	A
R	Load resistance	Ω
V_n	RMS voltage of the n-th harmonic	V
V_1	RMS voltage of the fundamental frequency	V
$T(k)$	Temperature at discrete time step k	K / °C
$P_{\text{dissipated}}$	Instantaneous power dissipation	W
C_{thermal}	Thermal capacitance	J/K
V_{in}	Input voltage to comparator block	V
$V_{\text{threshold}}$	FPGA trip threshold voltage	V
t_{trip}	Time elapsed before trip signal fires	s
N_{samples}	Consecutive over-threshold sample count	—
f_{sampling}	FPGA sampling frequency	Hz
n	Number of fractional bits (fixed-point)	—
V_{real}	Real-valued voltage before quantization	V
$V_{\text{fixed-point}}$	Quantized fixed-point voltage	LSB
$P(\text{Error} \mid \text{Pattern})$	Posterior fault probability	—
V_{limit}	Hardware trip threshold (40 V)	V
V_{current}	Present measured voltage	V
t_{prophecy}	Predicted time to failure	s
$\left.\frac{dV}{dt}\right\|_{\text{avg}}$	Average voltage slope over 500-sample buffer	V/s

5 Diagnostic Benchmarks and Safety Limits

The operating thresholds used throughout the framework were derived from published standards issued by ABB, Tesla, NASA PCOE, and IEC 61000-4-5, and were then calibrated against the specific characteristics of GaN and Germanium devices. Table 3 summarises the three operating zones and their associated system responses.

The normal operating band of 228–232 V is deliberately narrow. Keeping the window tight means that even modest deviations trigger the medium alert zone,

giving the diagnostic engine early visibility into developing faults before they become critical. The 35–40 V transient window for the critical trip was chosen because GaN breakdown typically initiates at the lower bound; the upper bound represents the absolute hard limit beyond which damage is certain.

Table 3: Operating Benchmarks and Safety Limits

Operating Zone	**Voltage Range**	**System Action**
Normal operation	228 V – 232 V	Stable load operation
Medium alert	21.6 V – 26.4 V ripple deviation	Early failure indicator logged
Critical trip	35 V – 40 V transient amplitude	Circuit disconnect triggered
Post-trip	—	Transition to Sleep Mode

Once the FPGA fires the critical trip, the system does not simply halt. It transitions to Sleep Mode, which parks the main power conversion path while keeping the diagnostic sensors alive. This is important — a post-fault snapshot of the sensor data is often the most valuable input for understanding what caused the trip in the first place.

6 Dashboard Launch Procedure

The interactive dashboard was intentionally kept simple to launch. Rather than requiring a separate Python environment, the visualisation code is pasted directly into MATLAB, keeping the entire workflow within the same toolchain used for simulation:

1. Open the Python text file included in the repository.
2. Copy the code block in full.
3. Paste it into the MATLAB Command Prompt and press Enter.
4. The dashboard will open, showing live *Spike Prophecy* alerts, component-level health scores, and the current operating zone for each monitored channel.

No additional dependencies or configuration are required beyond a standard MATLAB installation.

7 Conclusion

The central argument of this work is that EV power system protection needs to operate on two timescales simultaneously: nanosecond reflexes to prevent damage in the moment, and longer-horizon reasoning to anticipate failures before they happen. Bison-Prophet addresses both through a staged architecture where FPGA hardware handles the former and a Bayesian diagnostic model handles the latter.

The framework is intentionally platform-agnostic. While the simulations here were validated against the NXP S32G274A, the FPGA logic and Bayesian model

are not tied to any specific silicon. The most immediate next step is a hardware-in-the-loop deployment on the S32G274A itself, where the simulated trip latencies can be compared against measured values. Looking further ahead, the Bayesian model is well-suited to extension across multi-cell battery packs, where cell-to-cell variation creates exactly the kind of heterogeneous fault landscape that posterior inference handles well.

How to Cite

If you use this framework, dataset, or any part of this work in your own research or project, please credit the authors using the following citation:

APA Format:

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